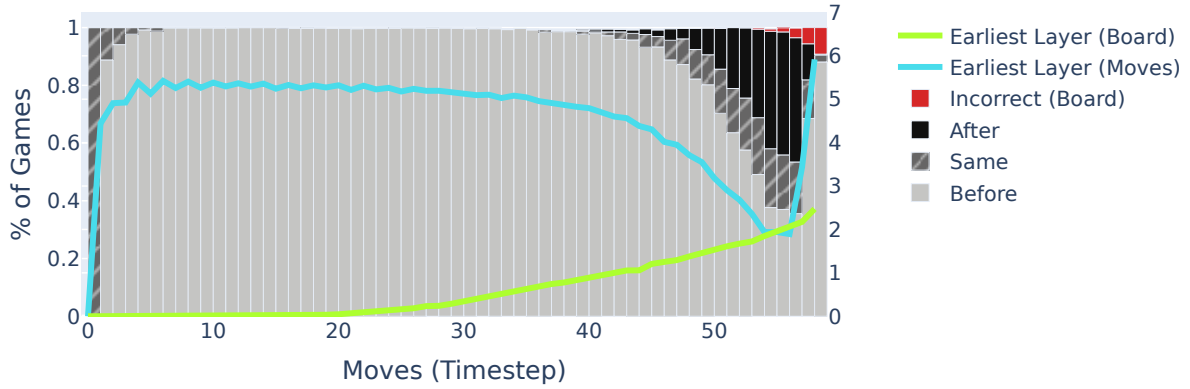

Figure 8: Percentage of times **90%** of the board state is computed before/after move predictions are made.

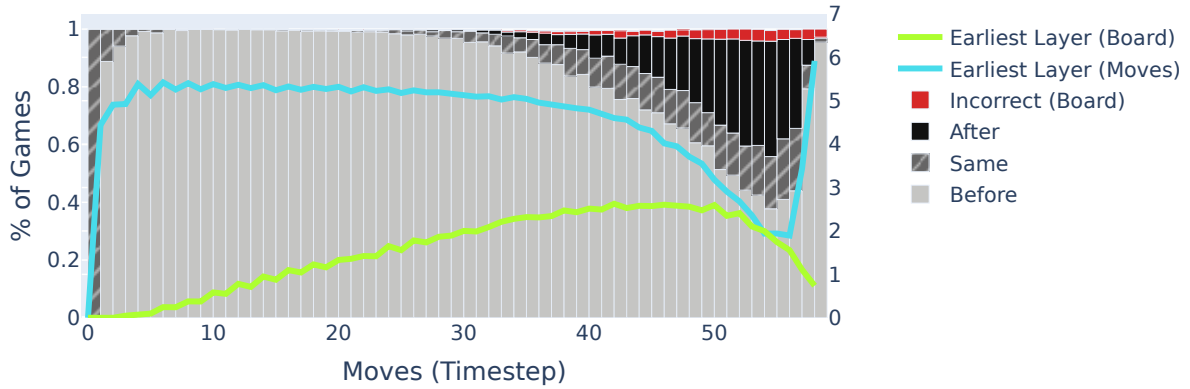

Figure 9: Percentage of times the **"minimum set"** of necessary board state is computed before/after move predictions are made.

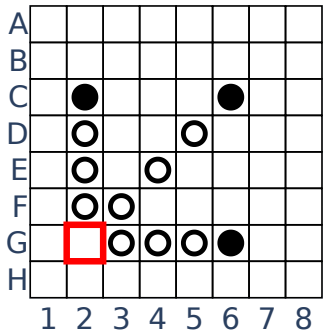

Figure 10: Example of "minimum set" of tiles that make move G2 legal.

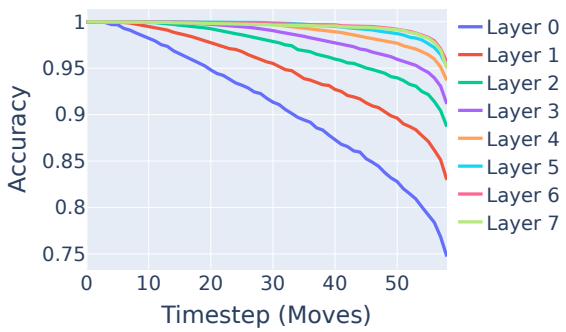

Figure 11: Accuracy per timestep for our linear probes.

successful interventions (Li et al., 2023b; Turner et al., 2023), linear probes are much easier to interpret, as they imply that features simply correspond to vectoral directions.

Another challenge is knowing which features to probe for, which can lead to pitfalls. Taking OthelloGPT as an example, classifying {BLACK, WHITE} versus {MINE, YOURS} leads to different takeaways, which illustrates the danger of *projecting our preconceptions*.

Speaking of incorrect takeaways, our last point concerns the expressivity of probe models. With an expressive-enough probe, there is a danger of the probe computing or memorizing the desired feature that one is looking for, rather than extracting (Pimentel et al., 2020a; Saphra and Lopez, 2019). Still, some researchers view linear classification

Baseline: Random	x^1	x^2	x^3	x^4	x^5	x^6	x^7	x^8
0.856	0.215	0.152	0.112	0.079	0.049	0.015	0.004	0.001

Table 5: Top-1 error rates when applying the unembedding layer to earlier layers. As a baseline we apply OthelloGPT's unembedding layer on a randomly initialized GPT model.

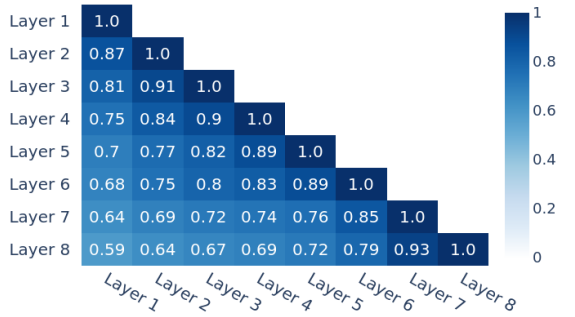

Figure 12: Cosine similarity scores between linear probes across layers.

as inadequate (Pimentel et al., 2020b; Saphra and Lopez, 2019). We view our work as evidence that linear probes do have interpretable and controllable power, and anticipate these findings to generalize to larger language models.